Final Project Proposal

CAI 5720 | Initial proposal

Due November 9, 2026, at 11:59 PM ET

Working project title: Does an early sepsis warning hold up at a hospital it has never seen?

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Research question and clinical significance

Using only data charted up to a given ICU hour, can a model flag adult ICU patients who will meet Sepsis-3 criteria within the next 6 hours, and does it stay accurate and calibrated at a hospital it was never trained on? The intended users are bedside nurses and rapid response teams deciding whether to start a sepsis workup. The unit of analysis is the patient-hour, the index time each ICU hour, and the horizon 6 hours. Earlier sepsis treatment improves survival, yet Wong et al., JAMA Internal Medicine 2021, found that a widely deployed commercial sepsis model performed far worse in external validation than its developers reported. We want to measure that transfer gap and what drives it.

Proposed dataset

We will use the PhysioNet/Computing in Cardiology Challenge 2019, Early Prediction of Sepsis from Clinical Data, version 1.0.0, at https://physionet.org/content/challenge-2019/1.0.0/, open access under CC BY 4.0 with no credentialing and described in Reyna et al., Critical Care Medicine 2020. It has 40,336 ICU patients and more than a million hourly rows: 20,336 from Beth Israel Deaconess with 8.8% sepsis and 20,000 from Emory with 5.7% sepsis. Each hour has 8 vital signs, 26 labs and 6 demographic or admission variables, with heavy lab missingness. The target is the hourly SepsisLabel, positive from 6 hours before Sepsis-3 onset; the positive class is sepsis.

Planned methods

We will develop on Beth Israel with patient-level cross-validation, keeping each patient's hours in one fold, and hold out all of Emory, untouched, until the model, predictor set and threshold are locked. Imputation, scaling and any recalibration are fit within development folds only; patient ID and anything recorded after the index time are excluded. Predictors are the latest values, short-term trends, and indicators of whether each lab has been measured so far. We will compare a training-prevalence reference and a SIRS-style rule with interpretable logistic regression and gradient boosted trees, each tuned by a small cross-validated search.

On Emory we will report AUROC, AUPRC as average precision, ROC and precision-recall curves, a calibration curve, Brier score, and a confusion matrix with sensitivity, specificity and PPV at a threshold set in development data from a fixed alert budget, plus alerts per patient-day and patient-level bootstrap confidence intervals. Interpretation uses standardized logistic coefficients, permutation importance, two contrasting held-out patients including a missed case, and errors by age group and sex with counts. A sensitivity analysis without the measurement indicators tests whether the model reads physiology or clinician ordering habits.

Expected outcome

We expect both learned models to lose discrimination and especially calibration at Emory, and part of their signal to reflect when labs were ordered rather than lab values. We will produce a reproducible Colab notebook and a report on how much performance drops at a new hospital and how many alerts a unit would face. Limits are charted-data labels, a retrospective design, two academic centers, and no race or treatment data. Still tentative: the alert budget level, whether to drop post-onset hours, whether to subsample hours per patient, and whether to also train on Emory and test on Beth Israel.